The Mountain Goats, Every Night

Scraping and modeling 34 years of live setlists from themountaingoats.fandom.com

The dataset

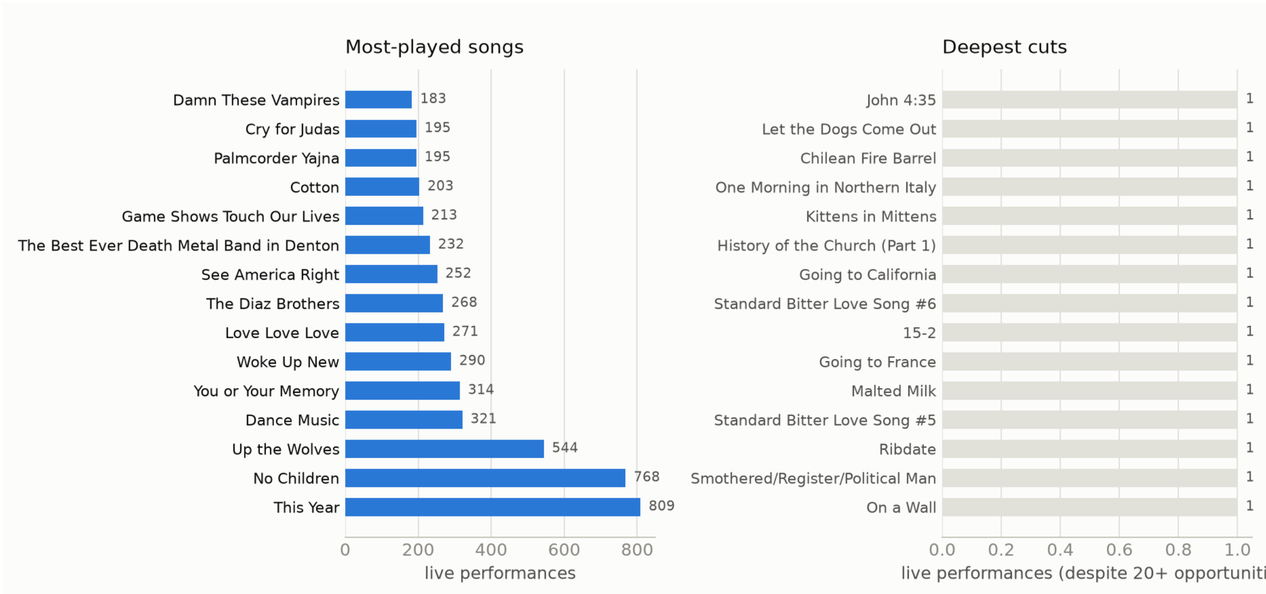
The wiki lists 1,579 live shows from 1992-05-31 through 2026-05-26 — solo John Darnielle sets, full-band tours, radio sessions, festival slots, and Extra Glenns/Extra Lens side-project shows. 1,317 of those have a documented setlist: 22,699 individual song performances across 845 distinct songs, spanning 97 named tours. 1,739 performances have at least one linked YouTube/Vimeo video.

Getting from a wiki category page to those numbers took a few real fixes:

- **Tour isn’t in the page text — it’s a wiki category** (e.g. *Peter Hughes Farewell Tour 2024*). Two earlier scraper attempts both missed this and scraped a null tour for every show.
- **Encores aren’t marked in the setlist table.** They’re stated in prose in the Notes section — “The encore was songs 19 through 23” — referencing the table’s printed order numbers. A small parser turns ~30 recurring phrasings of that sentence into a per-song encore number.
- **Song identity needed real normalization.** The wiki links each song to its own page, but the same song shows up under underscore-joined slugs, space-joined display text, and case variants of both. Left unmerged, this silently splits a song’s play count across rows — a real bug caught mid-project, where “Southwood Plantation Road” initially showed 1 play instead of its real count, 165.
- **Covers needed their own source of truth.** Sniffing setlist notes for the word “cover” only catches about a dozen of them — the wiki actually maintains its own Category:Covers, which tags 156 of the songs played live here. Every song carries an is_cover flag from that category, and the prediction model excludes covers from its candidate universe entirely.
- **A disambiguated title quietly duplicated as its own note.** 20 songs have their own wiki page distinct from a same-named album (Tallahassee the song vs. Tallahassee the album); the setlist link's visible text is just the song title, but its title attribute is the disambiguated form. The note-cleanup step was stripping the resolved title from the raw cell text instead of what was actually there, leaving the plain link text stranded as a bogus note on hundreds of performances. Fixed by stripping what's actually in the raw text.

The pipeline is two-stage: **fetch** downloads and locally caches every wiki page via the MediaWiki API (resumable, incremental on later runs), and **build** parses the cached HTML into tidy tables entirely offline.

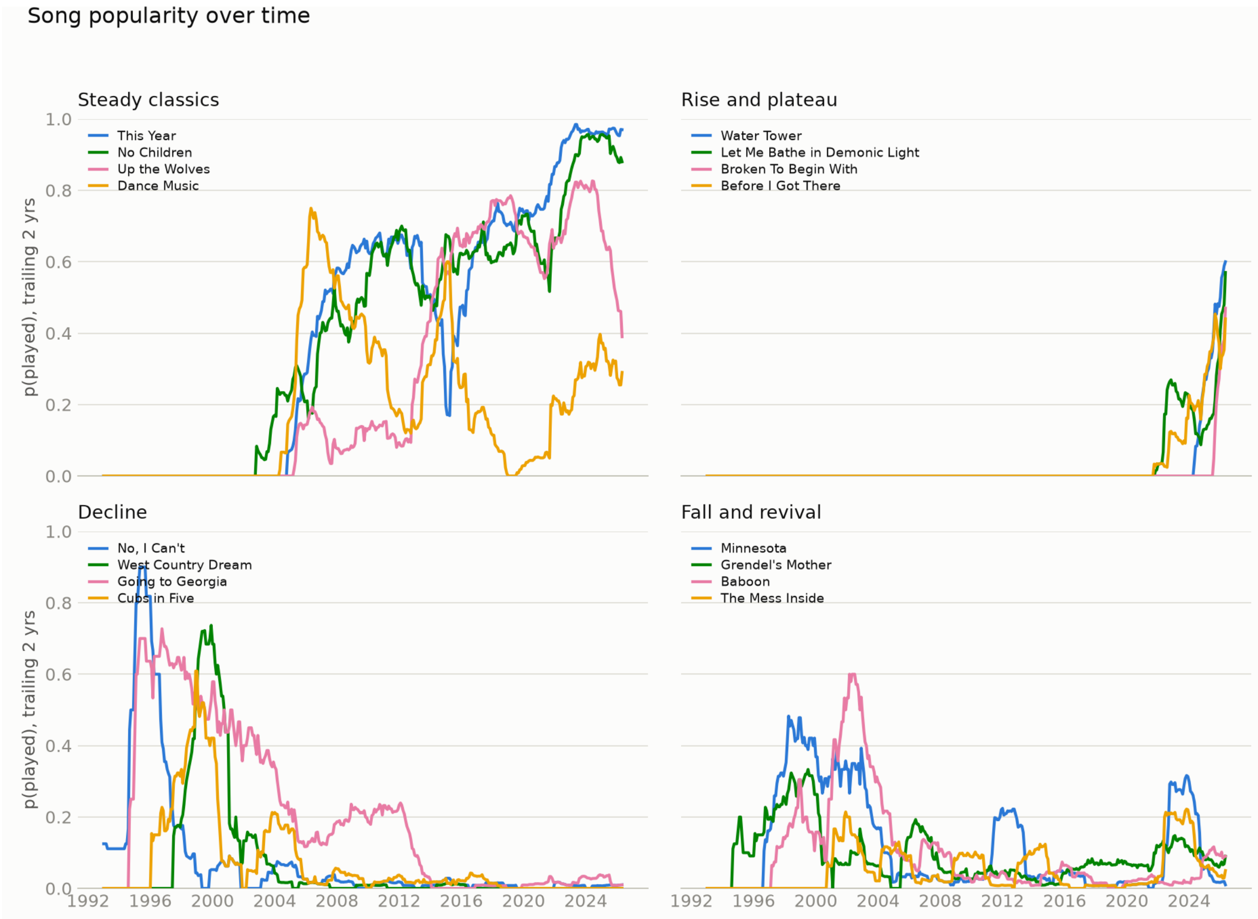
What gets played, and what doesn't



"This Year" and "No Children" anchor the top of the list — both played at a majority of all shows since entering rotation. The deep-cuts panel isn't just least-played-ever (most of the 845 songs have only 1–2 plays and aren't interesting on their own); it's the bottom of an opportunity-adjusted, shrinkage-smoothed play rate among songs with at least 20 chances to be played since their live debut — so a song only counts as a deep cut if it's been consistently passed over, not just recently written.

Song popularity over time

Treating each song's live history like a word's frequency in Google Ngrams — a trailing 2-year play rate sampled monthly — turns 34 years of setlists into trajectories: songs sit at zero before they're written, rise as they enter rotation, plateau, decline, and sometimes come back.



These four groups were picked automatically from the computed trajectories, not hand-selected: the most-played songs overall (steady classics); songs that debuted in the back half of the catalog's history and have climbed straight into rotation since (rise and plateau — a song appearing from nothing and taking hold); the biggest gap between early-career peak and recent play rate (decline); and songs with a real trough between two active periods (fall and revival).

The decline panel caught something the wiki itself corroborates: “Going to Georgia” was a fixture through the mid-2000s, then drops off hard — a 2012 show note has John Darnielle explaining, mid-show, that he considers it a “bullshit song” with an “asshole” narrator and doesn’t want to play it again. The data shows the falloff independent of that anecdote; the anecdote just explains it.

Predicting the setlist

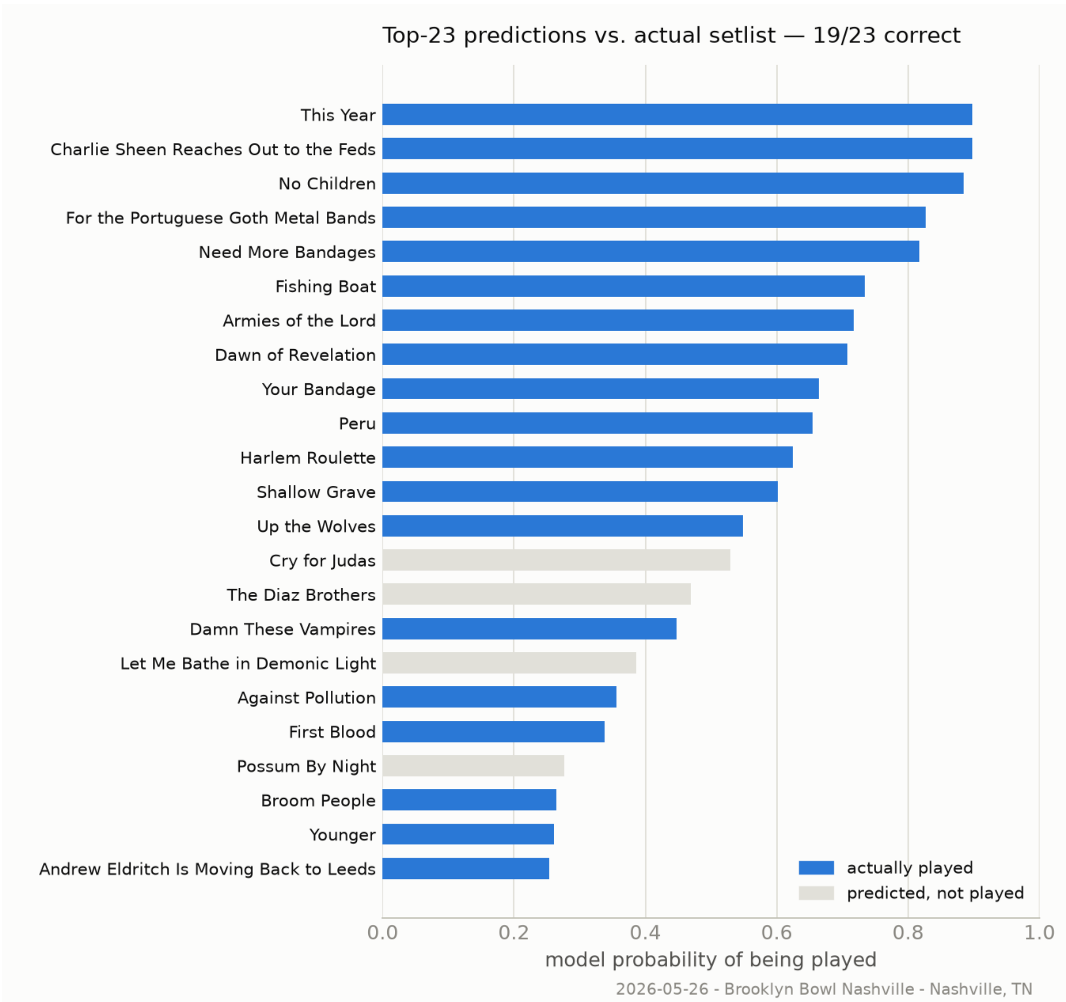
Framed as one binary outcome per (show, song) pair — is this the night song X gets played? — every one of the 1,310 dated, setlisted shows becomes a training example, using only information available before that show: how recently and how often the song's been played, how it's fared on the current tour, its age, and whether it's a shortened-set special appearance (radio/festival/TV).

A logistic regression on those features is compared against a simple but strong baseline — an exponentially-decayed play rate alone — on a strict temporal split (train through 2022, test on the 207 shows from 2023 onward):

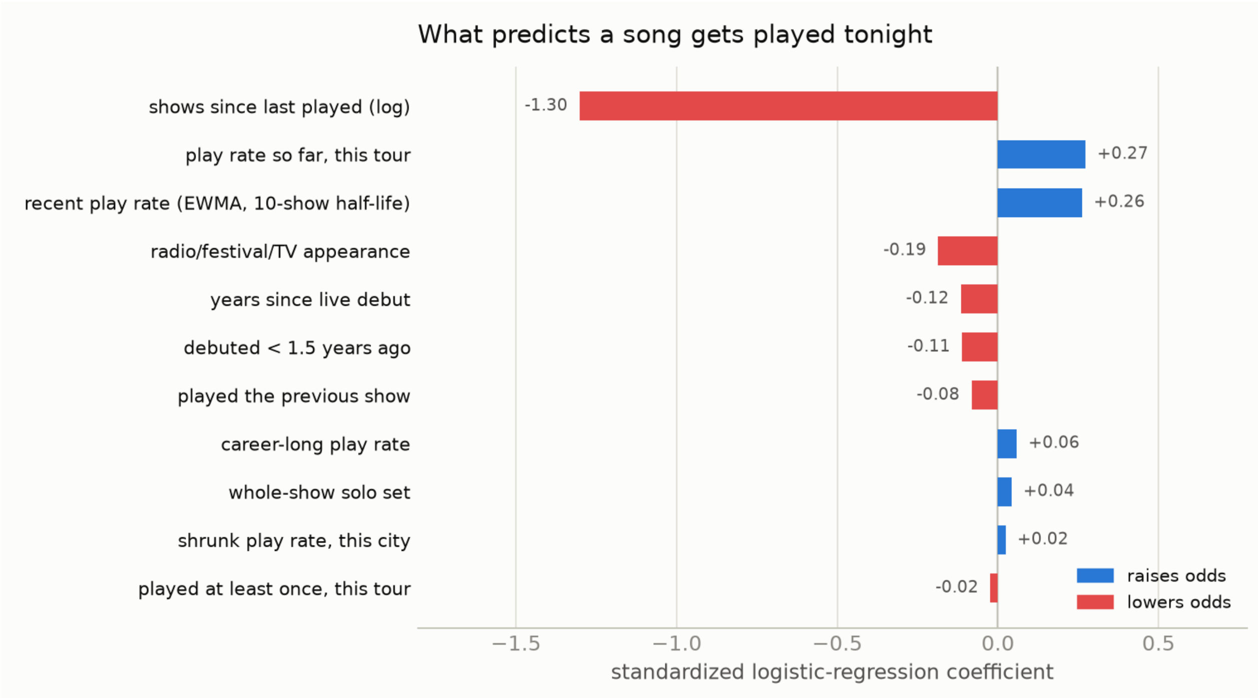
model	log-loss	Brier score	top-n setlist recovery

Baseline (decayed play rate)	0.0822	0.0195	51.1%
Logistic regression	**0.0673**	**0.0167**	**59.4%**

Top-n setlist recovery: for each show, take the model's n highest-probability songs, where n is the actual number played that night, and measure what fraction were right. On the most recent show in the data:



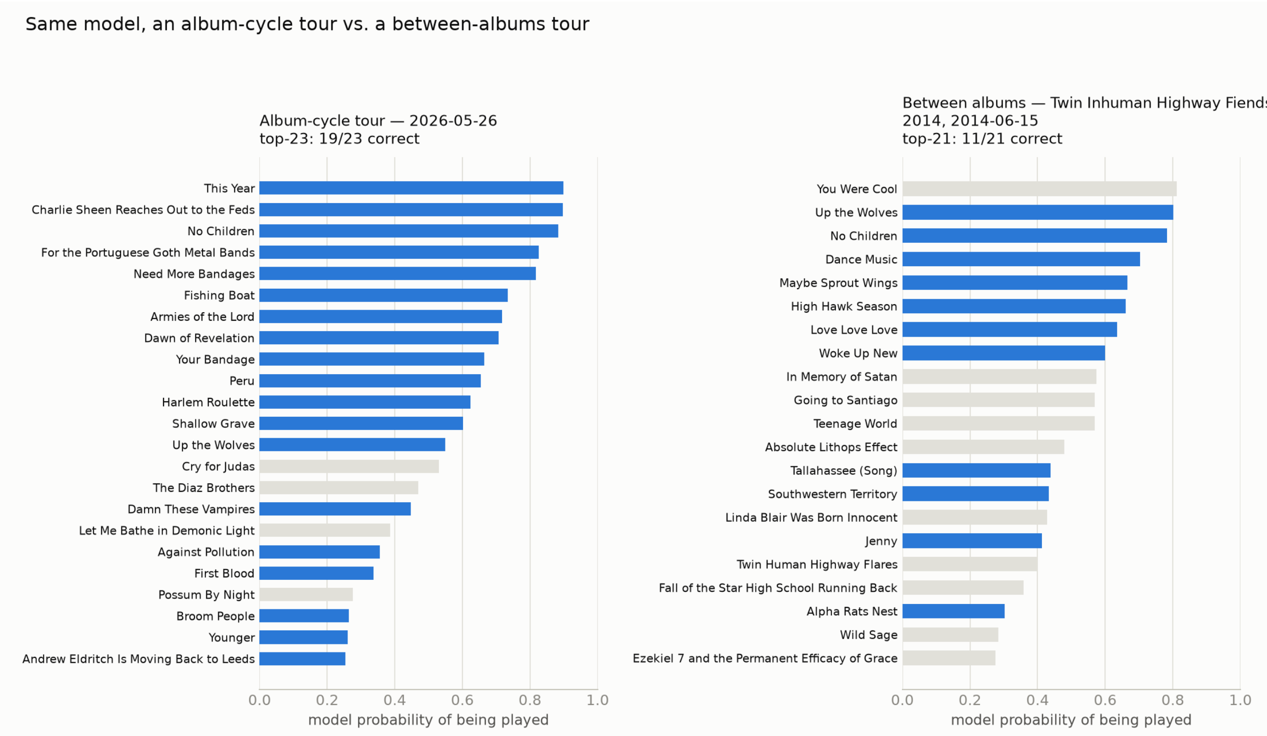
What actually predicts a setlist



Recency dominates everything else combined — if a song was played a few shows ago, it's overwhelmingly likely to come back soon. That's the visible mechanism behind the tour "rotation": John Darnielle appears to work from a live pool of songs and cycle through it, which is exactly what tour_rate (play rate so far, this tour) picking up an independent signal on top of recency confirms. Two smaller, genuinely interesting effects: conditional on recency, brand-new material is actually less sticky than catalog average, and radio/festival/TV appearances reliably favor hits over deep cuts. Two more recent additions, both real but modest: a whole-show solo set slightly raises a song's odds, and a shrunk city-level play rate (see Does geography matter? below) picks up a small independent "local favorite" effect.

Is this just tracking album hype?

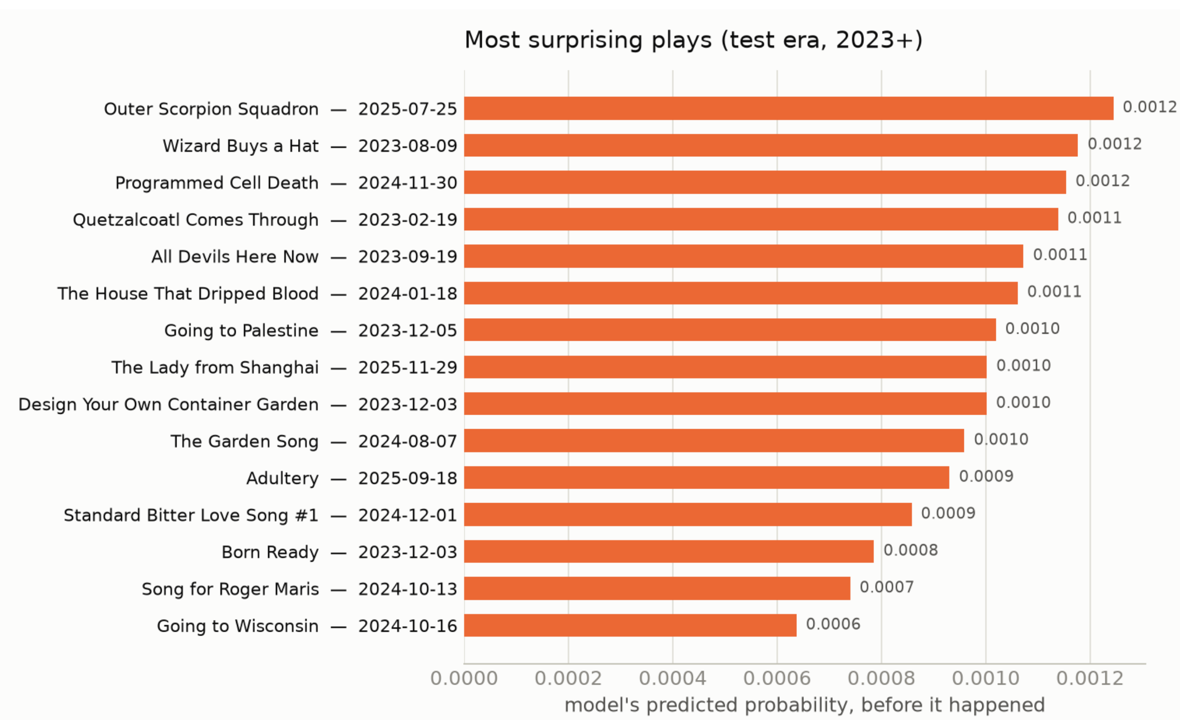
The 2026 example above is a new-album tour, and new_material is a real feature in the model — worth checking whether the strong recovery number is mostly "predict whatever's newest" in disguise. Among all 2014 tours with at least 8 shows, the one whose actually-played songs had the lowest share of new material — picked automatically, not by hand — is the Twin Inhuman Highway Fiends Tour 2014, squarely between Transcendental Youth (2012) and Beat the Champ (2015):



The between-albums tour is visibly harder to predict — 55.1% top-n recovery tour-wide, versus 59.4% for the full 2023+ test era — despite 2014 being in-sample (which should make it look artificially easier, not harder). Without a record to promote, the setlist draws more evenly across a wider pool of similarly-loved catalog songs, so there's genuinely more entropy to predict. The model isn't just riding hype; if anything, album cycles are the easy case.

When the model gets surprised

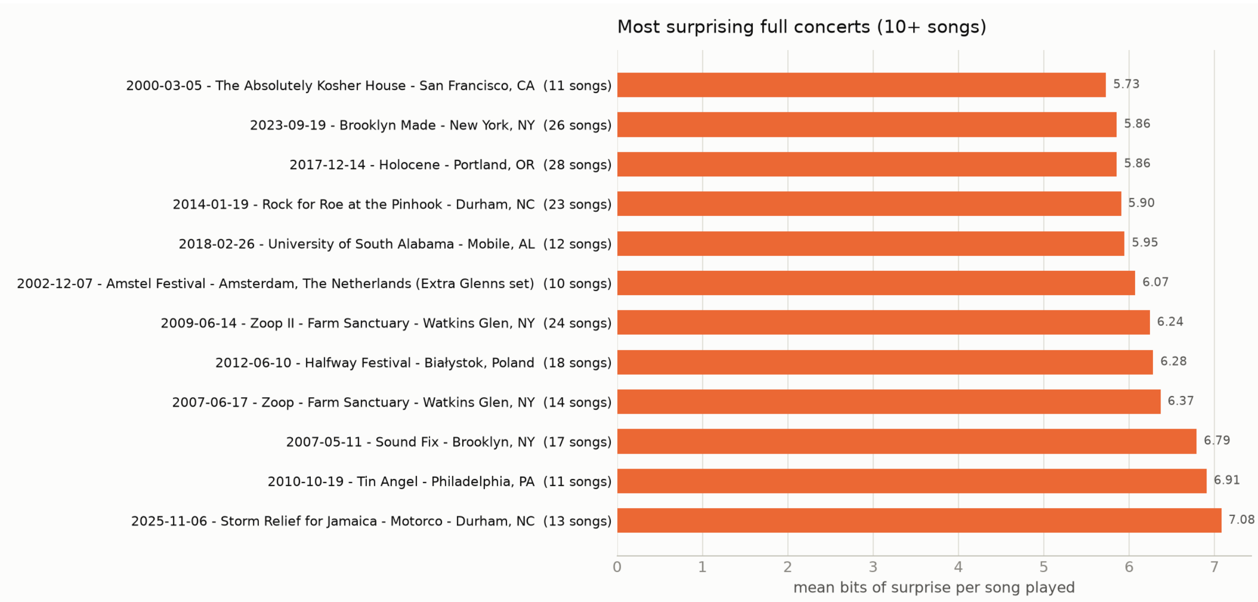
The flip side of a good model is a good list of its misses — nights the model gave a song almost no chance, and it got played anyway:



These are basically a machine-generated “rarities and one-offs” list: songs that came back from years of dormancy for a single night, often at a show with some specific occasion.

Most surprising concerts

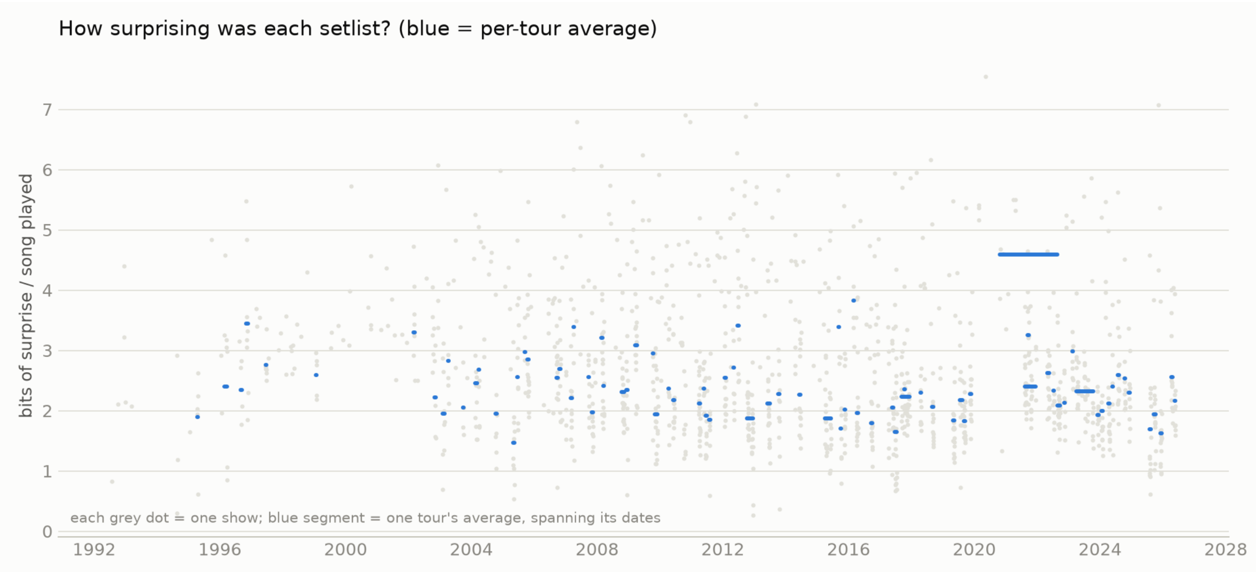
Individual surprising plays are one thing; whole surprising concerts are a different, more interesting question. Averaging each song's pre-show surprisal across a whole setlist answers this directly, filtered to real setlists (10+ songs) so a 1-song guest cameo can't trivially top the list:



These cluster hard around benefit shows, festivals, and untoured one-offs — a Hurricane Jamaica relief show, a “Rock for Roe” benefit, Merge Records' 35th-anniversary show, farm-sanctuary benefit gigs. Special-occasion shows pull deep cuts the regular tour rotation wouldn't produce, which is exactly what this metric should catch.

How surprising was each setlist as a whole?

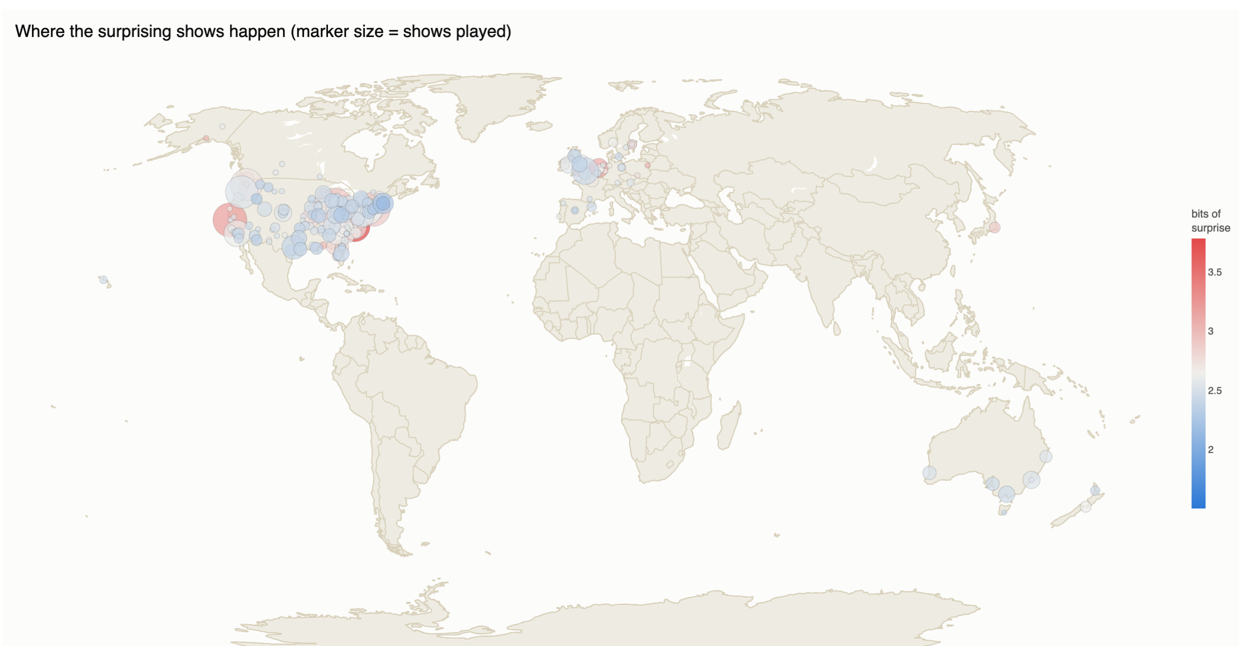
Zooming out further: the same pre-show probabilities give a surprise score for an entire night — the average bits of information across the songs actually played. Aggregated per tour (touring is bursty, so a fixed show-count window saws at tour boundaries):

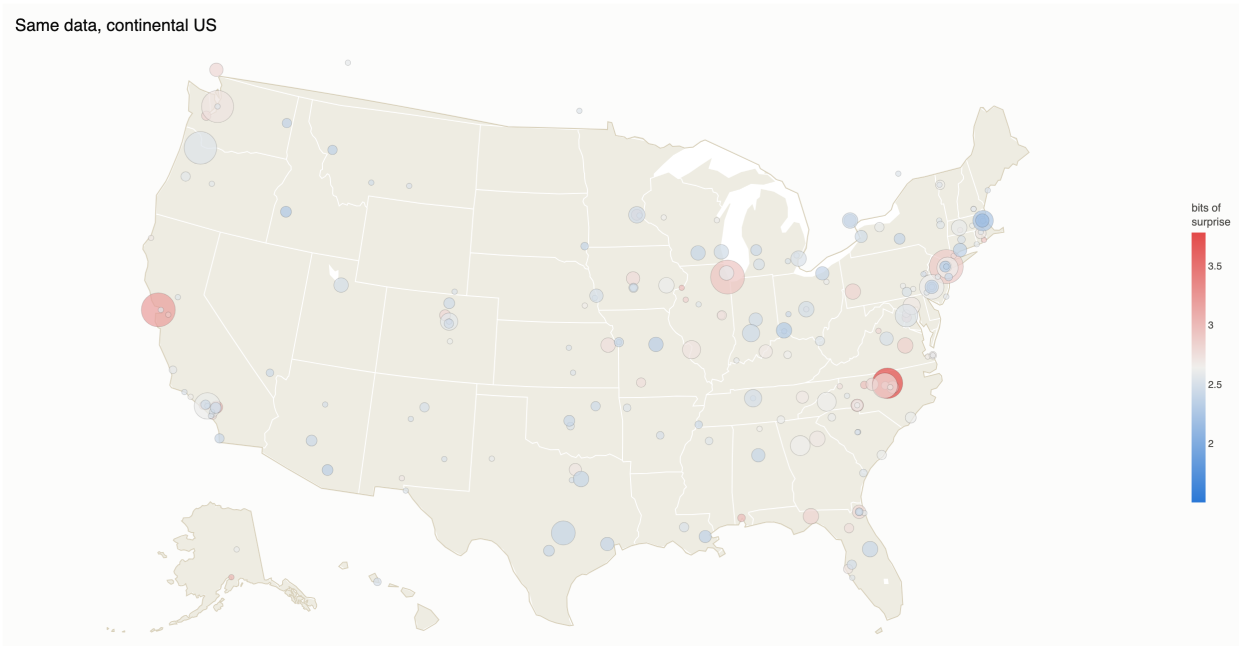


The tours with the highest average surprise are almost all solo, stripped-down tours — Winter Solo Tour 2024, the Ghost Cave Incubation Chamber solo non-tour, All Roads Lead to Lincoln Solo Mini-Tour — which is what motivated adding is_solo to the model above. Its coefficient came back small, though, meaning a whole show's unpredictability isn't fully explained by "solo" alone.

Does geography matter?

A live-music instinct worth checking against data: some cities feel like they get better, weirder, more surprising shows. San Francisco, in particular — is that real, or availability bias? The same per-show surprisal, grouped by city and empirical-Bayes shrunk toward the global mean (city-level data is thin: 312 cities, median 2 shows each), answers it:





San Francisco checks out: 3.20 shrunk bits against a 2.64 global average, rank 4 of 310 cities — and with 57 shows in the sample, shrinkage barely moves it off its raw average (3.28), so this isn't a small-sample fluke. But it's not even the strongest pattern: North Carolina dominates the top of the list — Durham (#1, 31 shows, 3.78 bits), Pittsboro (#3), Raleigh (#6) — and NC is the single most surprising region in the country. That's John Darnielle's home turf (he lives in Durham), and it tracks: hometown shows disproportionately include benefit gigs, record-release shows, and extended sets — the same kind of occasion that dominates the most-surprising-concerts list above. This motivated adding city_song_rate (a shrunk, causally-computed "does this song run hot in this city" rate) to the prediction model itself, not just this descriptive analysis — its coefficient is real but small, meaning local-favorite effects exist but are a minor correction on tour rotation, not a dominant force. Cities were geocoded against an offline database, not fabricated or hand-typed, matching 98.9% of setlisted shows.

Caveats

- The wiki is fan-maintained and lists most, not all, shows — coverage is noticeably better from the mid-2000s on.
- The model has no access to anything outside setlist history — no lyrics, no album themes, no explicit "songs currently being toured." tour_rate and song_age are proxies for that.
- Song identity is deduplicated via wiki link slugs where available; a handful of never-linked rarities may still have unmerged spelling variants.
- is_solo is deliberately conservative, not exhaustive: it only fires on an explicit "solo show" note or a tour branded "Solo," so it likely under-flags true solo shows the wiki didn't call out as exceptional — especially pre-2002, before a full-time backing band was the norm.
- City-level analyses only cover geocoded shows (98.9%) and are inherently thin for most cities (median 2 shows) — that's exactly what the shrinkage is for.

Reproducing this

```
PY=/Users/bdoughty/opt/miniconda3/envs/tmg-scrape/bin/python
$PY scrape.py fetch && $PY scrape.py build  # refresh the raw data
$PY analyze.py                # deep cuts, tour summaries
```

```
$PY predict.py               # the setlist model + surprisal
$PY geocode_cities.py       # offline city geocoding
$PY geography.py           # city/region surprisal, shrunk
$PY timeseries.py          # popularity-over-time series
$PY plot_report.py         # this report's figures
$PY plot_map.py           # the surprise maps
```

Full data dictionary in README.md; wiki-scraping gotchas and repo layout in CLAUDE.md.